

UNIVERSITI TEKNOLOGI MARA

**CHEST X-RAY DIAGNOSIS SYSTEM USING VISION-
LANGUAGE MODELS IN PICTURE ARCHIVING AND
COMMUNICATION**

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**Chest X-Ray Diagnosis System Using Vision-Language
Models in Picture Archiving And Communication**

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SUPERVISOR APPROVAL

Chest X-Ray Diagnosis System Using Vision-Language Models in Picture Archiving And Communication

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This thesis was prepared under the supervision of the project supervisor, Farah Syazwani Mohamed Rashid. It was submitted to the Faculty of Computer and Mathematical Sciences and was accepted in partial fulfilment of the requirement for the degree of Bachelor of Information Systems (Hons.) Intelligent Systems Engineering.

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JANUARY 27, 2026

STUDENT DECLARATION

I certify that this thesis and the project to which it refers is the product of my own work and that any idea or quotation from the work of other people, published or otherwise are fully acknowledged in accordance with the standard referring practice of the discipline.

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JANUARY 27, 2026

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LIST OF ABBREVIATION

CXR	Chest X-Ray
BERT	Bidirectional Encoder Representations from Transformers
GPT	Generative Pre-trained Transformer
ROGUE	Recall-Oriented Understudy for Gisting Evaluation
BLEU	Bilingual Evaluation Understudy
PACS	Picture Archiving Communication System
VLM	Vision Language Model
LLM	Large Language Model
CT	Computed Tomography
MRI	Magnetic Resonance Imaging
AI	Artificial Intelligence
ViT	Vision Transformer
SWIN	Swin Transformer
BART	Bidirectional encoding and autoregressive decoding

X-Ray-CoT	Chest X-Ray Chain-of-Thought
XMI	Extensible Markup Language
LLaMA	Large Language Model Meta AI
LoRA	Low-Rank Adaptation
PCP	Commonly Primary Care Physician
DICOM	Digital Imaging and Communications in Medicine
RIS	Radiology Information System
HL7	Health Level Seven
NIH	National Institutes of Health
OHIF	Open Health Imaging Foundation

CHAPTER 1

INTRODUCTION

This chapter provides the background and rationale for the study. It also gives detail of background study, the issue and problem that led to research such research question, research objectives, scopes, significant and expected outcome.

1.1 Background Study

Medical imaging has a significant role to play in clinical diagnosis and treatment planning in the era of globalization. The chest X-ray (CXR) is one of the most commonly used diagnostic systems to identify diseases like pneumonia, tuberculosis, and lung cancer, among others. The World Health Organization (WHO, 2023) reports that in the world, over 2 billion chest X-ray tests are done annually, which emphasizes the importance of radiology in health care systems of the population. A chest X-ray gives a two-dimensional image of the thoracic area where clinicians can determine the condition of the lungs and other diseases. The variation in tissue density is represented by the variation of shades of gray, and it is possible to distinguish anatomical structures and possible diseases (Imaging the Chest, 2001). Although it was used widely and has a diagnostic value, the rising number of chest X-ray tests has added pressure to radiologists.

A number of studies have observed that workloads and extended interpretation periods can be a cause of radiologist fatigue and inconsistencies in diagnostic errors. Stec et al. (2018) established that lengthy reading can result in diagnostic error, especially when the subtle findings exist. In the same way, Hopstaken et al. (1996) stressed that in order to diagnose pneumonia based on chest X-ray images, one should closely follow up. These difficulties give rise to the necessity of tools that will be able to help

radiologists and enhance the uniformity and decreasing the time spent on interpreting the results.

The Picture Archiving and Communication System (PACS) has been embraced by many healthcare institutions to handle an extensive number of medical images. PACS allows effective storage, retrieval, and viewing of medical images within hospital departments and has become an essential element of the current radiology practices (Arora and Mehta, 2014). Artificial intelligence (AI) is a supportive technology in the field of medical imaging that has been receiving attention in recent years. Past studies have demonstrated that AI-based solutions, which are Vision Language Model (VLM) and Large Language Model (LLM), can help discover the abnormalities and give explainable output and also place the most important cases in the focus of the radiologists (Singh et al., 2018).

However, although the current studies shows the potential of AI in interpreting chest X-rays and the significance of PACS in the radiology process, there is a lack of focus on the implementation of AI-aided interpretation into a PACS-like environment that recreates the real-world clinical practice. Also, there is no strong focus on offering interpretable and clear outputs that aid radiologists in the diagnostic process. The study will fill these gaps by showing an AI-enhanced diagnostic workflow in a simulated PACS setting, with a greater focus on assisting radiologists by enhancing their consistency, efficiency, and workflow integration.

1.2 Problem Statement

The process of manual interpretation of chest X-ray (CXR) images is very demanding in consistency, especially when it comes to the detection of subtle thoracic abnormalities. Radiologists may experience growing cognitive workloads that may impact negatively on reporting efficiency. Despite the use of artificial intelligence to aid in the interpretation of CXR, much of the available research has applied black-box prediction models, which offer weak descriptive or interpretable predictions. Lack of clinically meaningful explanations decreases the level of understanding and reliance clinicians give to explainable output (He, Wong, and Yoo 2025). Hence, it is necessary to investigate how pretrained vision-language models with the assistance of large language models can produce descriptive and interpretable outputs out of chest X-ray images to support radiologists better when interpreting images.

Picture Archiving and Communication System (PACS) has helped to greatly increase medical image storage and accessibility, but it does not meet the issues related to the interpretation workload or decision support when routine image review is to be conducted. A lot of AI systems that interpret chest X-rays include vision-language models and large language models, and they are typically built as non-integrated systems and do not work within the normal PACS workflow (Pham et al., 2025; Thawakar et al., 2024). Consequently, it needs to develop image interpretation based on VLM and LLM that can be successfully incorporated into a PACS-like image visualization, interpretation review, and reporting workflow.

There has been little emphasis on determining the functionality, system performance, workflow integration, and overall usability of VLM- and LLM-based systems as compared to the available methods. This raises doubts on the usefulness of such systems as diagnostic support tools. Thus, the proposed AI-assisted prototype has to be evaluated in terms of its technical functionality and system performance and integration of the workflow, and its results should be compared to the outcomes presented in the literature.

1.3 Research Question

Three research questions have been identified prior to conducting this research:

1. How effectively can a pretrained Vision–Language Model generate descriptive and interpretable outputs from chest X-ray images, and how can these outputs be presented within a simulated PACS environment?
2. How effective is the proposed system in integrating AI-generated image interpretations into a PACS-style workflow for demonstration purposes?
3. How do the generated outputs compare with existing AI-based chest X-ray interpretation approaches reported in the literature?

1.4 Research Objectives

Three research objectives need to be accomplished for this research:

1. To implement a pretrained Vision–Language Model for generating descriptive and interpretable textual outputs from chest X-ray images and demonstrate its integration with simulated PACS workflow
2. To develop a simulated PACS-style workflow that supports chest X-ray upload, image visualization, display of AI-generated interpretations, and report export.
3. To evaluate the proposed prototype in terms of technical functionality, system performance, and workflow integration by comparing it with related studies in the literature.

1.5 Research Scope

The research scope of this project is focused on developing the prototype PACS-integrated chest X-ray diagnosis system by applying the pretrained Vision-Language Model (VLM), which is SWIN-BART, to automate the process of the interpretation of medical images and the pretrained Large Language Model (LLM), which is LLaMA, for descriptive output from Hugging Face. The scope of this is limited to the 2D frontal chest X-ray image that has been received from Kaggle, which contains public datasets. The datasets contain at least 110k X-ray images that will be used for demonstration

purposes only for the system. The system workflow will allow the user to upload X-ray images; the images then will be processed and will be interpreted and analysed using pre-trained VLM and assisted by LLM to identify the disease. The project evaluation will include the technical functionality of the system, include metrics such as the stability of the system that can withstand image upload without crashing next. Does the system's pretrained model process the image and generate output without error? Next, evaluate system performance, including response time in seconds and consistency of output that generates the same output after uploading the same CXR image and comparing the existing AI integrated with the PACS environment, like including uploading images, image display, generating AI text, and exporting reports.

1.6 Research Significance

The research significance purpose of this study is to develop a prototype Chest X-ray (CXR) that integrates a pretrained Vision Language Model (VLM) and a pretrained Large Language Model (LLM) into a simulated PACS workflow. The function of the system is basically to automatically analyse the CXR image and give explainable output to assist the radiologist and medical worker in producing consistent output that is faster than they can interpret by themselves. This research contributes to the medical imaging and AI field by exactly presenting how explainable vision-language models and large language models can be applied in a PACS workflow environment for assisting in clinical decisions. This study focuses on the feasibility of using a public dataset rather than relying on a private patient dataset. The benefits of this research are basically for the radiologist and medical worker because the prototype can analyse and generate explainable output that is consistent and faster, which can reduce the cognitive workload. Healthcare also benefits because the prototype can improve workflow efficiency and reduce the reporting time. This also benefits the patient indirectly due to the faster diagnoses and lower human error. Finally, the prototype can benefit the medical student and trainee radiologist because it can be a platform that supports learning and diagnostic reasoning by combining image features with clinical explanations.

The research also supports the Sustainable Development Goal (SDGs), which is SDG 3, Good Health and Well-Being, where the prototype enables faster interpretation than manual, which is more consistent with explainable diagnostics. The system supports contributing to improved healthcare system performance, reducing diagnostic error, and better patient solutions. This also supporting the SDG 4 which is Quality Education that ensure the prototype with natural language explanation of the X-ray findings will serve the prototype as the support mechanism for the learning tool for medical trainees that profiting in enhance diagnostic skills and ensure self-directed learning.

1.7 Expected Outcomes

The expected output for this project is a list of descriptive and interpretable textual output that has been generated by a pretrained Vision-Language Model (VLM) and a pretrained Large Language Model based on the chest x-ray images, presenting the ability in foundation models for automated CXR interpretation. The model used is only for inference components, excluding training, to provide a medically meaningful explanation of identified abnormalities.

A simulated-style interface will be developed to assist in prototype demonstration and visualization. The interface will allow the user to upload an image, display the image for viewing or monitoring, and generate explanation output side by side to represent how the model outputs could be integrated into a radiology workflow. The interface's purpose is to demonstrate the PACS environment, like that used by hospitals, to demonstrate integration and visualization.

The system will be evaluated based on functional testing, system performance, and integration within the simulated PACS workflow. The technical functionality will evaluate if the image is successfully being processed by the pretrained model, producing output without error, and the stability of the system, such as not crashing

when uploading an image. Next is system performance, which is response time per second in generating output and consistent output generation with the same input. Finally, the workflow integration will be evaluated based on if generated output is present, enabling coherent interaction between image monitoring and also AI-generated interpretation.

CHAPTER 2

LITERATURE REVIEW

A literature review is a critical assessment of existing published research studies, articles, and academic sources on a given topic of study used to comprehend what is known, what constraints exist, and why new research is necessary.

2.1 Medical Image

Medical imaging, including fundamental elements of the modern healthcare system, allows every clinician to make observations of anatomical structures and physiological abnormalities without requiring any hard process. Computed tomography (CT), magnetic resonance imaging (MRI), radiography, and ultrasound contain knowledge that can assist in diagnosis, treatment planning, and monitoring the disease. Based on the modalities list, radiography medical image is the most reliable and widely used procedure because it has a lower cost, fast results, and requires fewer resources (Lu & Wang, 2025).

The growing digitalization of medical imaging has led to the increase of the medical data image that has been stored within the hospital information technology system. The dataset of the medical imaging contains visual and also clinical explanations that are combined into radiology reports and electronic health records. The usual clinical findings result from comparing the image, which varies based on its modality. In thoracic imaging, the patterns the clinician can observe are atelectasis, cardiomegaly, effusion, infiltration, mass, nodule, pneumonia, and pneumothorax. The medical image of CXR captures both the two-dimensional (2D) and also 3D capturing of the full human anatomy and pathology. (Lu & Wang, 2025).

2.1.1 Chest X-Ray

CXR, known as chest X-ray imaging, has been the most common anatomy and pathology examination that has been performed radiologically. Based on the Wang et al. (2017) statistic, almost 2 billion CXRs have been performed every year because of the cheap cost, high demand, and ability to identify thoracic diseases such as pneumonia, tuberculosis, cardiomegaly, pulmonary edema, and lung cancer. Even while CXRs are clinically significant, they can be difficult to interpret because of the slight differences in grayscale intensity, which might result in overlapping anatomical structures that could influence the harmful outcome. According to Wang et al. (2017), even for deep learning with strong training on big, large datasets, illness detection and recognition have become crucial tasks. Difficult inter-patient variability is a component of integrating; the noise of the pictures' variability, the images, and the existence of comorbidities are all components of integrating. Due to these difficulties, CXR interpretation has become the primary issue in the clinic challenges, particularly when it comes to the pool of CXR images that need to be analysed and published.



Figure 2.1 show the example of CXR image from NIH Chest X ray Dataset (Kaggle, 2025)

2.2 Clinical Challenges

The rapid growth in medical imaging volume has placed significant pressure on radiology departments, creating systemic bottlenecks in diagnostic workflows. According to Tanno et al. (2024), a global shortage of radiologists combined with increasing demand for imaging studies has led to heavier workloads, longer report turnaround times, and increased risk of diagnostic error. Radiologists are required to balance image interpretation with time-consuming manual report writing, which reduces efficiency and contributes to cognitive fatigue.

Additionally, substantial inter-observer variability exists in radiological interpretation, where different clinicians may reach different conclusions from the same image. This inconsistency can lead to diagnostic uncertainty, repeated examinations, and delayed clinical decisions. Tanno et al. (2024) further highlighted that both human-written and AI-generated reports may contain errors, emphasizing the need for assistive rather than fully autonomous systems. Overall, these challenges demonstrate that radiology workflows are constrained not only by image interpretation but also by reporting and documentation demands, highlighting the need for AI-assisted systems that support clinicians, reduce workload, and improve consistency while maintaining human oversight

2.2.1 Diagnostic Error

Radiology reporting is a time-consuming cognitive skill that involves abstracting visual observations into medical language. Islam et al. (2025) empirical studies have found manual report writing to be a key bottleneck in clinical processes, leading radiologists to spend less time interpreting images and more time on documentation. Free-text reports also result in different choices of terminology and writing style. However, identical findings may be described using different terminology by different clinicians, which makes it more challenging for downstream tasks to

perform data mining and quality auditing or provide clinical decision support. Tanno et al. (2024) noted that the lack of consistency in human reports leads to diagnostic uncertainty and reduces their value as training data for AI systems. Automated reporting aims to reduce these concerns by generating uniform interpretations of findings and impressions. However, creating clinically accurate and linguistically coherent reports involves modelling not only visual features but also medical language at considerable depth which motivates adoption of multimodal architectures.

2.3 Radiology Workflow

Today's radiology workflow is a complex multimodal chain with the PACS acting as the digital core for storage, access, and visualization of imaging informatics. It is the organized system for handling a variety of medical images (X-rays, CT, and MRI) with radiological reports serving as the official documentation of a clinician's analysis. This process is now under considerable stress with a worldwide shortage of radiologists and extremely high annual imaging data volume leading to diagnostic delay and owner's stress (Huemann, 2024; Islam, 2025; Ryu, 2025).

Comparative reading shows the essentially human-driven process of reviewing a patient's past images and reports to track disease progression or monitor treatment efficacy remains a fundamental step in this workflow. In interpretation, clinicians have a correct chain of thought ranging from recognizing abnormal visual features to formulating pathophysiology and constructing the final diagnosis. The manual creation of unstructured, textual reports at this stage is one of the main clinical bottlenecks, which can give rise to inter-observer variation and potential diagnostic error (Huemann, 2024; Ng et al., 2025; Islam et al., 2025).

2.3.1 Picture Archiving and Communication System (PACS)

PACS acts as the digital foundation of today's radiology departments, offering centralized storage, retrieval and distribution of medical images

such as CT scans, chest x-rays and MRIs. Wang et al. (2017) stated that PACS as a big data volume of imaging knowledge database since it collects large collective quantities of radiological studies including reports with its time and date. This integration provides the opportunity to follow-up patients longitudinally and allows for a continuity of care where both old image records can be compared with new ones. By the substitution of film archives, PACS brings tremendous benefits in terms of workflow efficiencies and speed on sharing image information among different hospital districts.

However, traditional PACS system were originally designed as a data tracking systems and not an analytical tool. They structure images based on metadata like patient ID, examination dates and modalities, but have zero knowledge of image content. A typical PACS does not interpret visual patterns, identify the pathological findings and draw inferences on clinical grounds from previously stored images. This makes it necessary for radiologists to perform detachable tasks such as detection of abnormalities, urgent case prioritisation and reading summary. This dependency on human interpretation is not scalable, especially in a high-throughput setting where thousands of imaging studies are accumulated daily.

In addition, PACS does not allow for automated detection of anomalies or decision support. It is not able to interpret suspicious areas, suggest possible diagnoses, or create initial reports. No priority given to images at storage level, irrespective of the clinical urgency. As a consequence, workflow is limited by the presence of readers specialized in the field and time delays may be experienced for urgent situations (emergency imaging). In this context, PACS is essentially a passive archive that stores and transmits imaging data but does not participate in the reasoning process for image interpretation.

These limitations expose a serious chasm between the availability and use of data in clinical radiology. PACS provides access to images, but does not turn the raw imaging data into useful clinical information. This has spurred studies that aim to incorporate intelligent systems, such as vision–language models and large language models, into PACS directly. In addition to the role of PACS being confined to that of merely providing a holding place for medical images, future applications of PACS may benefit from the inclusion of semantic capabilities and automatic interpretation support systems currently lacking.

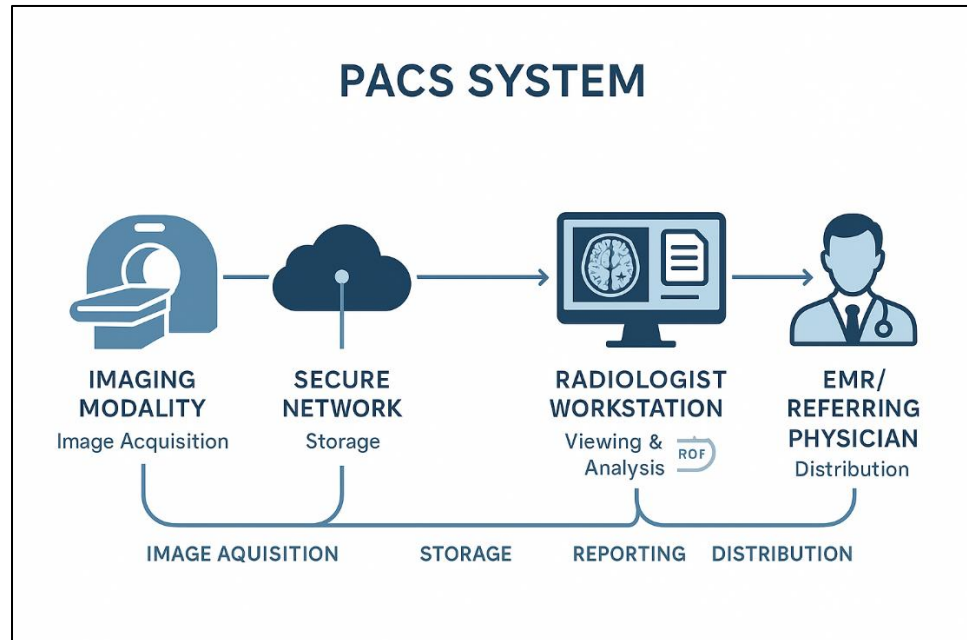


Figure 2.2 Example PACS system architecture from Curogram (Amoroso, 2025).

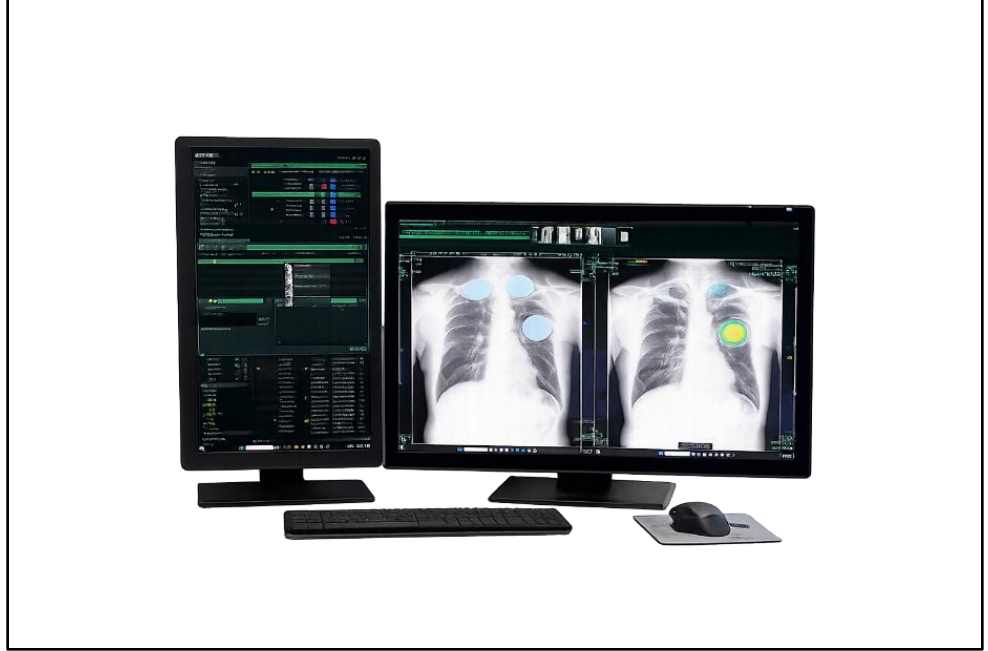


Figure 2.3: The example of PACS system interface (FUJIFILM Healthcare Americas Corporation, 2025).

2.3.2 System Integration

More recent studies have proposed integrating AI systems into the Picture Archiving and Communication System (PACS) to be able to analyse medical images in real time or close to it upon the capture and storage of an image (Wang et al., 2025). As opposed to the PACS being an archive and retrieval-only service, the proposed approach turns it into a smart processing space whereby image data are automatically analysed once they are stored.

Vision-Language Models (VLM) are the focus of this integration and are used to project chest X-ray images to uniform higher-level semantic features. Such representations contain clinically important visual data, such as anatomy, pathology patterns, and spatial data. These representations are then inputted into Large Language Models (LLM) that give explainable output in the form of structured textual interpretations to assist in bridging

the gap between image perception and clinical reporting. Wang et al. (2025) also suggested governance-ready models that were aimed at safe implementation in clinical workflows. Their model uses an abstention mechanism, which enables the system to make no predictions when there is low confidence or when the inputs are out of the training distribution. This minimizes the chances of giving erroneous results as the proven diagnoses.

Moreover, the framework has a human-in-the-loop design, according to which automated outcomes can be checked and confirmed by clinicians until a final decision is made. This will guarantee that medical professionals will be left in ultimate clinical decision-making even though they will still enjoy the use of computational support. The routine and time-consuming processes can be automated with the incorporation of AI into PACS. These are chest X-ray recognition, indexing and retrieval of disease tagging, and generating of preliminary draft reports. Such processes can be automated to lessen the amount of clerical work and enhance documentation reliability.

Notably, these systems do not mean that radiologists should be replaced. They, instead, act as decision-support tools. Through its capability to ensure clinician supervision and the implementation of clear safety technologies, AI-enhanced PACS allows a more scalable, more efficient, and safer radiology workflow. Figure 2.3 shows the simulation of AI in the PACS environment and shows where the AI platform, such as the vision language model and large language model, can be implemented in the PACS viewer.

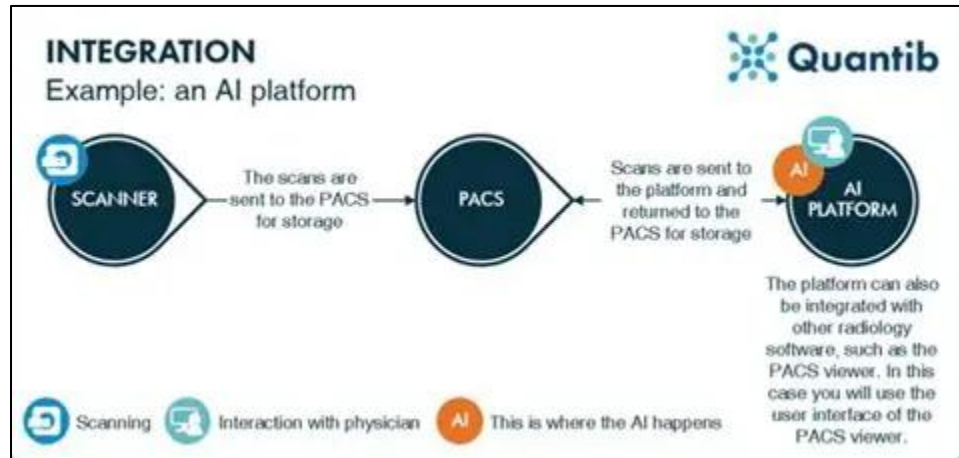


Figure 2.4 shows the explanation of how PACS integration with AI happens in the workflow. (Quantib - DeepHealth.,2025)

2.4 Image Captioning

Medical image captioning is a branch of automated textual image-related textual description generation used to provide a summary of clinically relevant information in radiographic images. In the case of chest X-rays, these involve identifying anatomical structures and identifying pathologies as well as characterizing their locations in accordance with the known medical science. Wang et al. (2017) demonstrated that large-scale datasets of chest X-ray images with radiologist-written reports can be used to teach models associations between visual patterns and clinical words under weak supervision, and report generation can be considered a vision-language, not a simple classification, task.

Medical image captioning is more accurate and must be done in line with clinical principles than general image captioning. Created descriptions should be clear, give a description of findings and anatomical location, and should not be ambiguous. Newer methods have thus employed transformer-based vision-language systems that combine visual and language data and generate multi-sentence outputs that are similar to reports written by radiologists and not single labels.

It was shown that transformer-based models are able to produce consistent and semantically relevant radiology reports, such as descriptions of multiple abnormalities observed within a single image (Islam et al., 2025). On the same note, Tanno et al. (2024) discovered that paired chest X-ray images and reports from large-scale vision-language models generated clinically consistent narratives equivalent to standard human-generated reports.

Recent works as well have focused on interpretability in medical image captioning. Ng et al. (2025) demonstrated that semantic grounding is enhanced by including chain-of-thought reasoning, and generated reports are more associated with clinical reasoning. In general, medical image captioning has become an important part of AI-assisted radiology, which helps to use the standardized reporting and decreases the amount of documentation.

2.4.1 Pretrained Vision Language Model

Pretrained vision–language models are ever more crucial in medical image analysis as they facilitate knowledge transfer from large-scale datasets to domain-specific tasks with sparse annotated medical data. In radiology, the acquisition and annotation of images, as well as image report generation, is expensive since experts must be included. Vision-language pretrained models mitigate the issue by learning visual and linguistic representations jointly from large-scale image-text datasets and transferring them to various downstream tasks, including medical image captioning and automated reporting generation (Wang et al., 2022; Huang et al., 2021).

Islam et al. (2025) investigated various vision encoders (Vision Transformer ViT and SWIN Transformer), paired with language decoders (GPT-2 and BART). ViT adopts over image patches to capture long-range dependencies, but however, it may miss detailed local information for

identifying subtle pathological patterns in CXRs. In comparison, the SWIN Transformer introduces a hierarchical window-based attention mechanism that keeps local spatial details while gradually accumulating global context, which is more fit for medical imaging applications with both discriminative and semantic visual recognition.

On the generation side, GPT-2 and BART model differently. While GPT-2 is an autoregressive language model that excels at generating fluent text, its lack of bidirectionality can constrain the ability to condition on complex visual features. BART, conversely, is a sequence-to-sequence transformer pre-trained with noise-corrupting objectives that ought to make it better at generating well-structured and contextually coherent text. This is important, especially for radiology reports, which should adhere to a well-established clinical structure and preserve the logic among sentences.

Islam et al. (2025) systematically studied several encoder–decoder architectures, inasmuch as reviewing the performance of ViT-BART, ViT-GPT-2, SWIN-GPT-2, and SWIN-BART for textual overlap (ROUGE), n-gram precision (BLEU), and semantic consistency (BERTScore). Overall, SWIN-BART produced the best results among all baselines, including a ROUGE-1 F1 of 0.4134, a ROUGE-L of F1 0.2935, a BLEU of 0.0648, and the highest BERTScore F1 of 0.8899 on our test sets. These results suggest that SWIN-BART can produce more semantically consistent results with reference to clinical texts and better linguistic coherence than ViT- and GPT-2-based systems.

The superior performance of SWIN-BART is due to the complementary nature of its component parts. The hierarchical visual representation learned by the SWIN encoder helps the model to localize fine-grained abnormalities in chest X-rays, while the BART decoder can generate clinically relevant and coherent language. Inspired by the above

comparative results, SWIN-BART is chosen in this work as the pretrained model for CXR report generation. Pretrained weights enable the model to utilize prior knowledge acquired from large-scale datasets, thereby reducing the dependence on deep domain-specific annotation and making automated radiology reporting faster and more accurate.

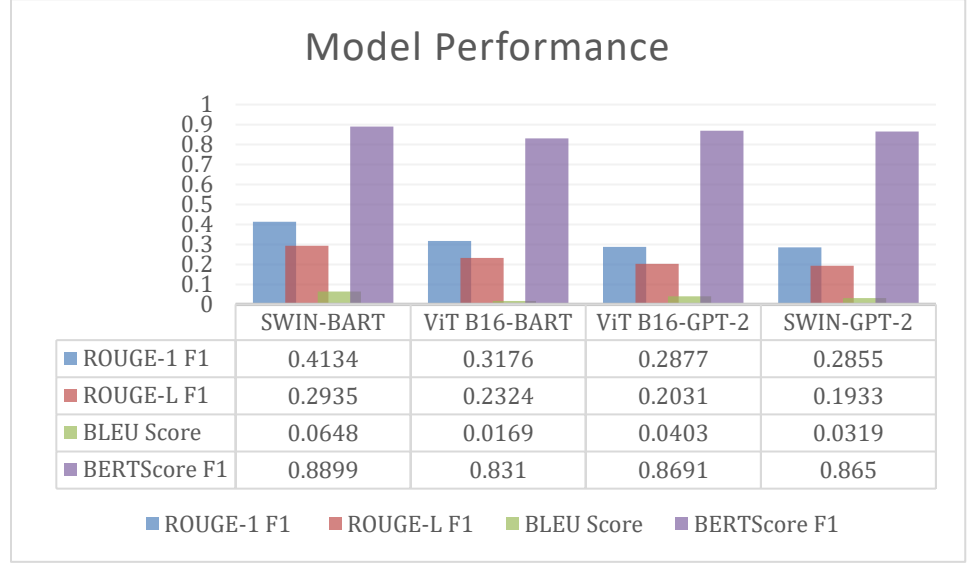


Figure 2.5 shows the comparison of model performance through different main technology dimensions. (Islam et al., 2025).

I. SWIN-BART

SWIN-BART is a vision-language transformer that processes multi-modal inputs to exploit the latent cross-modality feature representations for automated chest X-ray interpretation by translating radiographic visual contents to structured textual texts. The architecture combines a SWIN Transformer as the visual encoder and BART as the sequence-to-sequence language decoder by means of cross-attention so that output sentences are in images (Islam et al., 2025). The hierarchical architecture and shifted window attention of the SWIN Transformer are conducive to

capturing discriminative local pathological patterns as well as global anatomical context for chest radiograph analysis.

The IU-Xray results indicated that SWIN-BART generates semantically and syntactically coherent X-ray image descriptions. The model obtained a BERTScore F1 of 0.8899 and a ROUGE-1 F1 of 0.4134, reflecting high alignment between generated outputs and human-made descriptions (Islam et al., 2025). Moreover, the qualitative explanation also reveals that the RawLoss is capable of interpreting several anomalies from one X-ray image, thus proving its efficacy towards the X-ray resolution task.

Even though SWIN-BART performs well, however, it is constrained by fixed-length output production and the diversity of training data. Thus, these limitations could limit its capacity to capture complex or unusual disease presentations. Since the 2D vision-based interpretations are dependable, SWIN-BART does not natively support higher-level reasoning or explainable diagnostic logic, which therefore requires large language models for downstream reasoning and explanation.

2.5 Explainable output

Interpretability in medical imaging seeks to generate textual narratives that rationalize and explain predictions of the model as radiologists communicate their diagnostic reasons. Explainable algorithms do more than just make a final decision or label and generate structured interpretations that describe the observed abnormality and its clinical implications. Initial works formulated this problem as a vision–language task that requires mapping visual concepts from medical images into meaningful natural language rationales (Jing et al., 2017). This takes disease classification a step even

further since it models the descriptive and interpretative reasoning that radiologists use for image-based reporting.

Recently, explainable medical AI has seen significant advancement due to the Transformer-based vision–language architectures with encoder-decoder design. In such systems, visual encoders derive semantic information from image content, and language decoders generate textual explanations conditioned on that rich set of visual cues. Islam et al. (2025) showed that models of this nature are capable of generating fluent and semantically valid descriptions for chest X-ray images, detailing multiple health issues through coherent medical terms. This ability is further bolstered by large-scale vision-language models that can be trained on paired data of chest X-rays and radiology reports to generate clinically reasonable rationales approximating expert-level reasoning (Tanno et al., 2024).

However, despite these lines of progress, it still remains difficult to create explainable output that is visually and linguistically precise. Models must suppress hallucinations, are required to properly connect visual evidence with anatomy, and should communicate their results in a standardized clinical language. Ng et al. (2025) stressed the necessity of interpretability through reasoning-driven schemes that explicitly reveal intermediate inference steps, a process that precedes reaching ultimate conclusions. In general, explainable output is a key application of large language models in medical imaging and has high potential to boost transparency, clinician trust and decision-making and requires system design with caution for patient safety and clinical reliability.

2.5.1 Pretrained Large Language Model

Pretrained Large Language Models (LLMs) are becoming critical for medical image interpretation by allowing systems to produce structured textual explanations from visual information. Compared to vision-only classifiers, LLM can convert image features into clinically interpretable

natural language by including anatomical context and medical explanation. Recent studies have moved from prediction-only models to reasoning-centric ones, where the intent is to mimic a radiologist’s diagnostic pathway and thus enhance transparency and trust in AI systems.

Ng et al. (2025) introduced X-Ray-CoT, which uses a Chain-of-Thought prompting approach to lead an LLM through intermediate inferential steps such as identification of abnormalities, pathophysiological reasoning, and generation of a justified diagnosis. This approach yielded a balanced accuracy of 80.52% and an F1 score of 78.65%, better than standard black-box vision transformers, proving that explainability can be provided without sacrificing predictive performance. In line with this work, DeVide incorporates medical knowledge into pretraining and achieved a top-1 accuracy of 82.3% on zero-shot chest X-ray classification (Luo, 2024), which provides additional evidence for the capability of pretrained language models to perform generalizable and knowledge-aware medical reasoning.

Given that there is a growing need for XMI in the medical domain, of all pretrained LLMs available, LLaMA has very good generative performance and works well with reasoning-based prompting strategies. Analogous to smaller autoregressive decoders tuned primarily for fluency, LLaMA is a flexible large-scale language generation model, which can maintain logical coherence over multi-push explanations and be conditioned to generate structured reasoning that respects clinical workflows. Also, its open-weight architecture makes it more transparent and reproducible than closed-source counterparts. Given these considerations and its verified performance in CoT settings (Ng, 2025), the present experiment uses LLaMA as the pretrained language backbone for providing interpretable, clinically graspable explanation outputs.

I. LLaMA

LLaMA (Large Language Model Meta AI) is a Transformer-based Large Language Model (LLM) that acts as a central language and reasoning element for medical Vision-Language Models (VLMs). In applications in the medical domain, the LLaMA can be easily adjusted by further pre-training continued on biomedical literature and fine-tuning under tutoring to align outputs with professional clinical terminology/standards (Li et al., 2025; Ryu et al., 2025; Ng et al., 2025). This allows the model to produce clinically interpretable information from visual or structured inputs.

This has been shown in recent frameworks, where LLaMA is very effective for explainable diagnostic reasoning when guided with Chain-of-Thought prompting. The X-Ray-CoT framework demonstrated that LLaMA behaves like a radiologist when it can generate intermediate inference steps and then generate the final diagnosis to obtain an 80.52% balanced accuracy and a 78.65% F1 (Ng et al., 2025). In explicitly presenting reasoning steps, LLaMA-based systems increase transparency and trust for a clinician over black box vision models.

LLaMA has good reasoning power but is quite resource intensive in terms of training and inference. To address this, parameter-efficient adaptation strategies like LoRA and quantization have been used to generate smaller deployable versions that could be deemed acceptable for hospital settings (Ng et al., 2025; Wang et al., 2025). In summary, LLaMA offers a good trade-off between interpretability and performance, making it viable as the language backbone.

2.6 Related Works

The related work of this literature review is referred from existing works and clinical systems that enable benchmarks for comparison performance, result analysis, and limitations of the prototype that the new study is to fulfill.

2.6.1 Performance

Recent literature has highlighted that response time and integration speed are the most important performance measurements in artificial intelligence systems incorporated into PACS (Picture Archiving and Communication System), especially for emergency as well as high-volume radiology workstations. Joy et al. (2024) said healthcare organizations deploying PACS frameworks with AI algorithms were able to process images and analyze results up to 31% faster than by using a manual approach. For AI-enabled triage systems in emergency situations, detection of critical findings such as pneumothorax was at seconds from the time of image acquisition. The ability to reduce waits of several hours while waiting for radiologist review (having the urgent cases prioritized) and total turnaround time was substantial (Joy et al., 2024).

Besides the response time, diagnostic consistency was observed to be a key performance gain of AI and PACS integration. Jot et al. (2024) demonstrated that using a standardized AI interpretation model had a 25% reduction in inter-reader variability among the radiologists. Moreover, diagnostic consistency measurements went up to 91% for successfully installed systems, which means repeat reads of the same image delivered stable and reproducible results across diverse clinical settings. This repeatability is especially relevant in current radiology practice, where increased workload and tiredness may result in non-uniform human interpretation. Joy et al. (2024) pointed to the almost 49% of radiologists

with significant burnout, indicating that AI systems can be supportive in offering reliable initial readings.

In total, the conclusions indicated in Joy et al. (2024) show that the integration of AI into PACS is possible to significantly enhance responding time, consistency in diagnosis, and efficiency in the time worked by the staff. The study also emphasizes the necessity of strong technical infrastructure and security considerations in order to provide a steady and efficient system performance. These findings suggest AI-imbedded PACS systems may provide significant performance benefits in comparison with traditional workflow paths, but their implementation needs to be carefully coordinated to ensure the safety and security of clinical practices.

2.6.2 Result analysis

Analysis of results is compared against earlier PACS generations and current PACS that show a shift from passive storage to active diagnostic support in networked image repositories. The first related work, PACS, by Demner-Fushman et al. (2016), is described as a system with the main purpose of retrieving and distributing radiology examinations, in which images are stored and available for recall by radiologists to facilitate their interpretation but not used computationally during the diagnostic process. In contrast, current AI-facilitated PACS is entering an artificially intelligent workflow where images are being interpreted automatically after they are taken with fully automated interpretation of preliminary measurements or pathologies even before the radiologist opens the study. This results in a “head-start” effect where AI can contribute to the diagnostic pipeline by preparing the case for review rather than being merely a storage system, moving the workflow of routine radiology from a digital filing cabinet to a human-machine partnership (Demner-Fushman et al., 2016). However, the gap of clinical acceptance is significant, since many hospitals are still working with PACS that are not AI-ready because

of the technical difficulty of deploying AI solutions in a remote reading PCP using non-standardized DICOM tags differing from one institution to another. These obstacles to integration hinder the capacity for deployment and the operational potency that AI can have in day-to-day usage. Indeed, when AI systems are deployed from the sketch to the simulation of a real hospital environment for clinical use, their operation often suffers because of different data formats and workflow conditions and systems interoperability. This illustrates that, as visionary as their conceptual flow is, current implementations of modern logical flows still need maturity before they can be spread worldwide (Nguyen et al., 2021).

2.6.3 Limitations

Though the recent literature proves the effectiveness of deep learning and Vision-Language Models in the interpretation of the chest X-ray, there are a few limitations to the current research, which are echoed in the presented scope of the study. First, most of the existing literature concentrates on the accuracy of models or the classification, and little emphasis is placed on systems-level integration in clinical processes e.g. within PACS environments. Because of this, it is not clear how many of the proposed models would be practically implemented in a real radiology environment. The workflow simulation is covered in this study, but it is only on a simplified PACS-style setting and without complete hospital integrations like Radiology Information Systems (RIS) and HL7 communication standards. Second, the system generates descriptive and non-diagnostic outputs in order to prevent clinical decision-making. As a result, the resulting reports cannot be truly used in the actual diagnostics and can be inadequate in terms of depth and reliability to be used in the clinical interpretation. This restriction is indicative of the general challenge to current literature on safe usage of generative AI in medical fields. Third, the proposed system is limited to functional testing, system response, and

intergration workflow in the evaluation. There is no assessment of quantitative clinical performance indicators, e.g. sensitivity or specificity. The limitation can be linked to current research which places more emphasis on technical feasibility and explanation rather than direct diagnostic validation. Lastly, the system uses publicly available datasets of chest X-rays and they may not be representative of the diversity and complexity of real-world clinical data. Differences in imaging procedures, patient characteristics, and disease occurrence are thus not exhaustively covered in this research work.

CHAPTER 3

RESEARCH METHODOLOGY

The study uses a system development method to design a prototype used in the interpretation of chest X-rays in a simulated PACS-like process. The system combines a set of pretrained vision-language models and large language models to help analyse images automatically and generate a report. Examples of how the prototype is assessed regarding its feasibility as an AI-assisted diagnostic support system include functionality, performance, and integrating workflow. The methodology will also show how such a system can help improve efficiency and consistency and fit within the normal radiology workflows.

3.1 Methodology Table

Research methodology table 3.1 containing the objectives of the project, phases, activities, and the deliverables.

Objectives	Phase	Activities	Deliverables
To implement a pretrained Vision–Language Model for generating descriptive and interpretable textual outputs from chest X-ray images and demonstrate its integration with	Preliminary Study	Review literature related to chest X-ray diagnosis, Vision-Language Models, Large Language Models and PACS systems.	Background of Study, Problem statement, research objectives, research questions, research scope, and literature review

simulated PACS workflow		- Analyse the current problems of the related study. - Formulate research problem, objectives, and research questions.	
	Data Understanding	- Identify and select a suitable public Chest X-ray dataset from Kaggle website - Review dataset structure and labels	Selected Kaggle Chest X-Ray dataset and dataset description.
	Data Preprocessing	- Inspect images for format consistency - Resize images to ensure compatibility with the prototype system - Apply basic normalization where required	Prepared chest X-ray images ready for inference and system demonstration

		for stable inference and visualization	
	System Design	• Design system architecture (input–process–output) • Define the role of pre-trained Swin-BART VLM as a black-box image-to-text inference component • Define the role of the LLaMA LLM for generating explanatory like output generation • Design inference and output flow within a simulated PACS context	system architecture diagram, inference workflow design.
	System Development	• Configure and utilise pre-	Functional model with AI-generated descriptive

		trained VLM and LLM via API-based inference (no modification or training) • Design prompt templates for structured, non-diagnostic descriptive outputs • Implement the multimodal inference pipeline	text outputs
To develop a simulated PACS-style workflow that supports chest X-ray upload, image visualization, display of AI-generated interpretations, and report export.	System Implementation	• Develop a web-based frontend using Streamlit or Gradio for chest X-ray upload, visualization, and report display. • Implement backend inference services using FastAPI.	Streamlit/Gradio-based PACS-style viewer, FastAPI backend services, chest X-ray visualization interface, AI-generated diagnostic report module, MongoDB database, Dockerized system

		• Integrate pydicom for medical image handling and OHIF Viewer-style image visualization. • Store images and reports using MongoDB. • Deliver by using Docker.	
To evaluate the proposed prototype in terms of technical functionality, system performance, and workflow integration by comparing it with related studies in the literature.	Prototype Testing and Evaluation	• Conduct technical functionality testing of the prototype using representative CXR datasets by self-testing. • Measure system performance in terms of consistent output and response time • Evaluate workflow integration	Technical functionality testing results of the proposed prototype, system performance, integration workflow.

	Documentation	Documenting and Writing the report	Final Year Project report and complete documentation
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3.2 Preliminary Study

The preliminary analysis of the system concentrates on laying a solid theoretical and technical basis for the proposed system. Background Review A survey of the literature is performed regarding chest X-ray interpretation, Vision–Language Models (VLM), Large Language Models (LLM), and Picture Archiving and Communication Systems (PACS). This review intends to bring attention to the shortcomings of manual radiograph interpretation, such as diagnostic latency, inter-observer variation, and insufficient number of radiologists. Also, insights from already applied AI-based diagnostic systems are reviewed for recognizing some of the common obstacles regarding explainable solutions and their partial introduction to clinical routine. The review emphasizes that several existing systems are not integrated within the hospital information infrastructure, thus limiting their practical applicability to real settings.

Finally, the literature indicates a research gap in systems combining visual perception and natural language reasoning embedded with a PACS-like workflow. The integration is necessary to allow image organization, visualization, and automatic textual output creation within a common framework. The articles are collated from our arXiv, IEEE eXplore, Pubmed and Scopus. The findings in this stage are used to develop the background of the study, problem statement, objectives of research, questions of research, and scope of research. This phase will be completed with the establishment of a planned path for the later research phases.

3.3 Data Understanding

The data understanding stage involves the determination of an appropriate public chest X-ray dataset for demonstration and evaluation purposes. The NIH Chest X-ray dataset from Kaggle is utilized in this paper. The dataset includes 112,120 frontal-view chest X-ray images with disease labels. The dataset link is [NIH Chest X-rays](#)

The dataset was chosen because it is a large and public dataset, widely used in the history of chest x-ray study, and suitable for benchmarking and system demonstration with our method. With labeled images, one can also look inside the image content and see which disease categories are being diagnosed.

This part involves the exploration of dataset structure to comprehend image format, labeling methodology, and how the dataset is arranged. The provided sample chest X-rays and corresponding disease labels are reviewed for appropriateness for further preprocessing and system integration. Images and labels of chest X-ray radiographs are shown in Figure 3.1 and in Figure 3.2 respectively. This will result in a curated and well-characterized set of chest x-ray image datasets to be utilized as the main input for model inference and system demonstration towards later stages of this work.



Figure 3.1 shows the example picture of the X-ray in the dataset.

A		B		C	D	E	F	G	H	I
1	Image Index	Finding Labels		Follow-up #	Patient ID	Patient Age	Patient GeView	Posit	OriginalIm	Height
2	00000001_000.png	Cardiomegaly		0	1	58 M	PA		2682	2
3	00000001_001.png	Cardiomegaly Emphysema		1	1	58 M	PA		2894	2
4	00000001_002.png	Cardiomegaly Effusion		2	1	58 M	PA		2500	2
5	00000002_000.png	No Finding		0	2	81 M	PA		2500	2
6	00000003_000.png	Hernia		0	3	81 F	PA		2582	2
7	00000003_001.png	Hernia		1	3	74 F	PA		2500	2
8	00000003_002.png	Hernia		2	3	75 F	PA		2048	2
9	00000003_003.png	Hernia Infiltration		3	3	76 F	PA		2698	2
10	00000003_004.png	Hernia		4	3	77 F	PA		2500	2
11	00000003_005.png	Hernia		5	3	78 F	PA		2686	2
12	00000003_006.png	Hernia		6	3	79 F	PA		2992	2
13	00000003_007.png	Hernia		7	3	80 F	PA		2582	2
14	00000004_000.png	Mass Nodule		0	4	82 M	AP		2500	2

Figure 3.2: shows the example label, which refers to the disease of each picture in the dataset.

3.4 Data Preparation

The data preprocessing stage transforms the chest X-ray images for inference and for system demonstration only. Since the dataset is already collected, preprocessing will be more concerned with image management rather than manual cleaning. All images are initially screened for matching format and resolution. The images are resized to fit the input format of the Vision–Language Model (SWIN-BART) as per the inference pipeline. Simple normalization is used for stability of inference and to increase visualization quality in the system interface. In contrast to standard deep learning pipelines, there is no model training and data augmentation in this work, as the system depends only on pretrained models for image-to-text inference. The end result of this phase is a collection of calibrated chest X-ray images that are conformal to the inference pipeline and PACS-like viewing, as well as systems demonstration.

3.5 System Design

The conceptual architecture of the proposed chest X-ray interpretation system is shaped by the system design phase. The system adopts an input–process–output architecture to distinguish the functional roles of the components. Input The input to the system is chest X-ray images submitted by users via a PACS-style interface. We split the processing stage into two major modules: a Vision–Language Model (VLM) and a Large Language Model (LLM). We instantiate Swin-BART as a vision-language model, and it is used as a black-box inference module. It generates visual features from chest X-ray images and preliminary image-conditioned textual representations. We choose Swin-BART because of its transformer-based visual encoder and because it has the ability to align image features with language tokens. Since Swin-BART generates no fully structured explanatory reports, we propose a Large Language Model for Model-Agnostic Interpretation (LLaMA) to interpret them. The outputs of Swin-BART are fed into the LLaMA model using structured prompt templates, which allows for textual reports in a descriptive and interpretable document (findings and impression-style summaries). This modular structure allows the system to go beyond

simple image classification in terms of generating human-understandable descriptions based on visual inputs and enables transparent interpretability as the volcanic phenomena are better described.

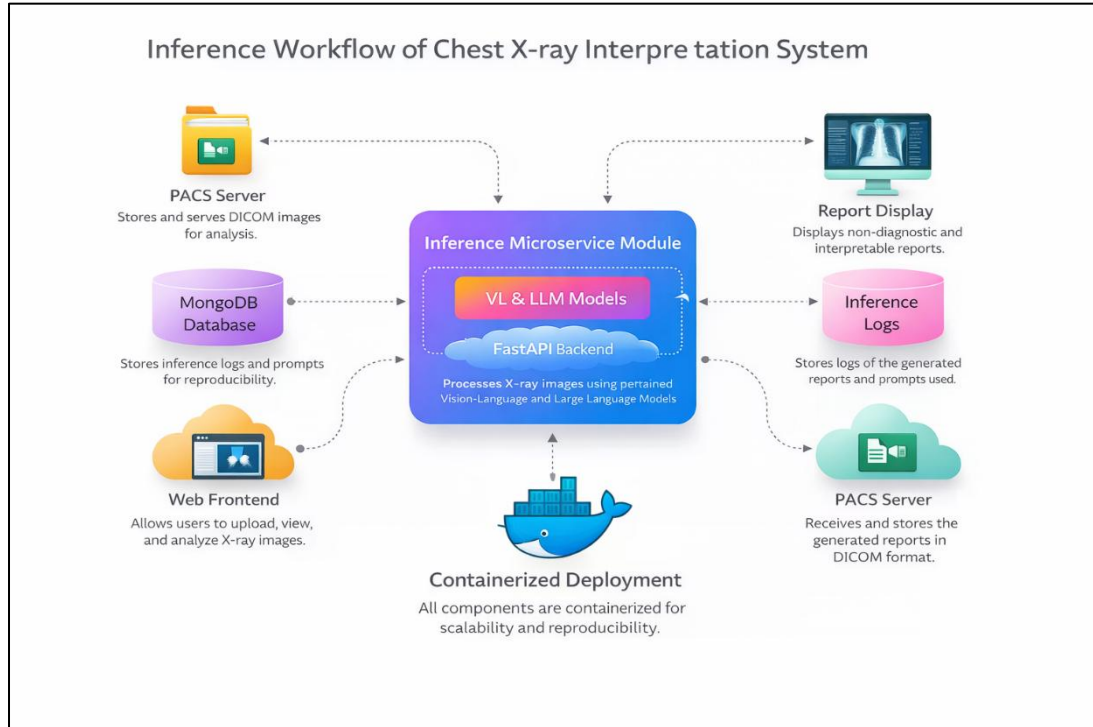


Figure 3.3 shows the inference workflow of the chest X-ray interpretation system.

Based on the figure 3,3 The workflow of the inference starts by uploading an image of a chest X-ray via the Internet-based PACS-type frontend. The uploaded photo is sent to the inference microservice module, which is the hub of computations of the system. In this module, the FastAPI server receives the picture and processes it with the necessary preprocessing and sends it to the Vision-Language and Large Language Models. The Vision-Language Model takes the image of the chest X-ray and works on it to create the visual features and provide a first image-conditioned textual representation. This output is further programmed with a prearranged template prompt and fed into the Large Language Model to be further interpreted. The Large Language Model generates the input into a refined and readable written text, i.e. observations and summary sections. In the process, the generated text and prompts as a part of

inference-related information may be saved in the MongoDB database and inference logs to ensure that the process can be reproduced and analyzed. The text output generated is fed back to the frontend where it is reflected with the associated chest X-ray image in a report the display interface. The complete pipeline of inferences is implemented in a containerized system to provide portability and reproducibility of the system. The workflow is a simplified model of a clinical setting even though it mimics a PACS server to which an image can be stored and retrieved. The workflow in general shows that a combination of the user interface and the interaction with backend services and multimodal AI models is necessary to turn the images of the chest X-rays into descriptive textual representations.

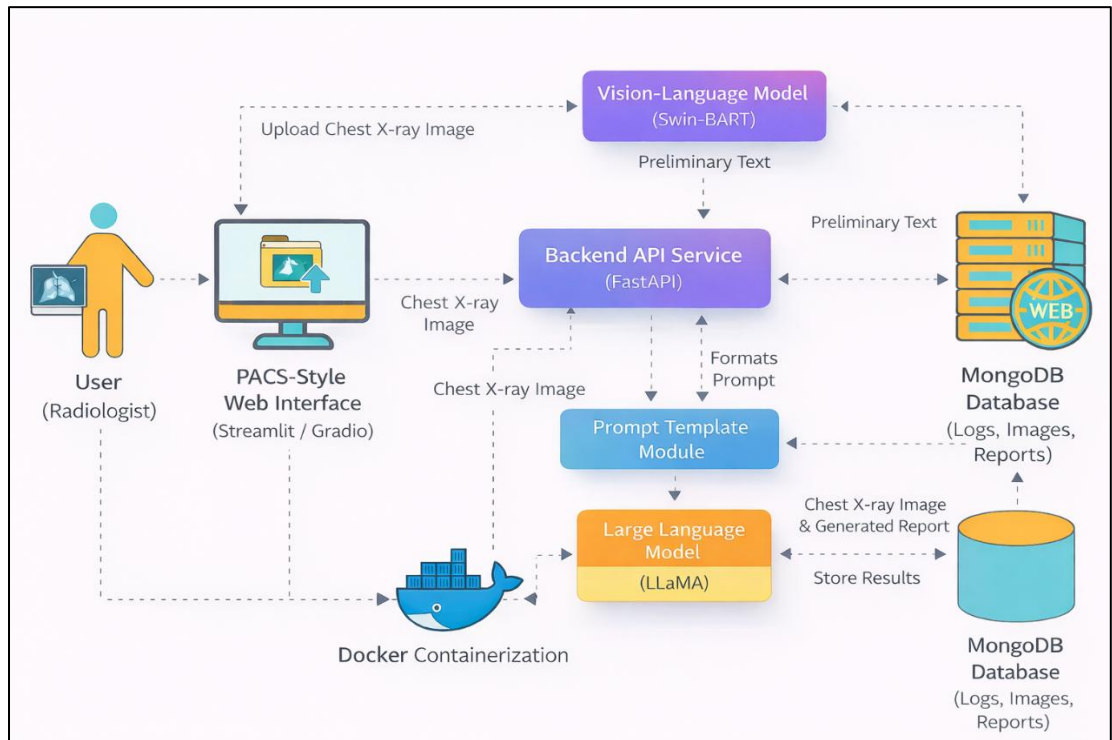


Figure 3.4 shows the system architecture diagram of the integrated PACS prototype.

The system architecture aims at providing the interpretation of the chest X-rays through the modular service-based system architecture. A radiologist (user) can interface with the system at the user level with a PACS-like web interface, developed

with the help of Streamlit or Gradio. In this interface, the user is able to post the chest X-ray images and access the images and textual output it will generate. The PACS-style interface interacts with the Backend API Service, which is established based on FastAPI. The backend is the key control unit of the system that receives images that are uploaded and also coordinates the processing activities and controls communication between various modules of the system. The Vision-Language Model (Swin-BART) is implemented within the backend processing layer and it analyses the input chest X-ray image and produces a preliminary image-conditioned textual representation. This output is sent on to the Prompt Template Module that structures the text into a form of a prompt that can be interpreted through language. The structured prompt is then sent to the Large Language Model (LLaMA) that can be an interpretative element to produce descriptive and interpretable textual reports, including observation and summary sections. The textual output thus generated is sent back to the service on the back-end to be processed. The database of the system is MongoDB that is used to store uploaded pictures, generated reports and the inference logs to be tested and analyzed. This storage element is decoupled with the processing modules, in order to preserve modularity as well as data management efficiency. Every part of the system, such as the front end, the back end, the model services, and database are all deployed in Docker containers to promote portability, reproducibility, and deployment ease in various environments. Such a design enables every component to work on its own and have co-ordinated communication with one another via defined interfaces. In general, the architecture facilitates a PACS-like workflow by combining user interaction, model inference, and data storage to form an inseparable system, and is maintained, scalable, and modular.

3.6 System Development

The system development phase moves from a conceptual design to an operational multimodal inference pipeline. We use both pre-trained Swin-BART and LLaMA models without any fine-tuning or retraining. This design choice has been made with the motivation that the system developed must be implemented within the time

available without any need for high-performance computing resources. Swin-BART, which is Swin transformers, has the ability to receive chest X-ray images and output human-like visual-language representations that depict the visual content of radiographs. These are generation results of textual descriptions conditioned on images based on visual features learned by the model. The generated prompts from Swin-BART are then filled with a structured prompt mimicry template and fed into the LLaMA model. The prompt templates are structured so as to help provide particle-based, non-diagnostic descriptions in the form of observations and summary statements, without explicitly making diagnostic decisions. This framework guarantees that the produced text is still descriptive and readable, instead of diagnostic. Through this integration, Swin-BART and LLaMA together constitute one integrated inference engine that converts raw chest X-ray images into meaningful textual interpretations. The pipeline progresses sequentially, including image ingestion, visual feature representation extraction, language-based query processing, and report generation. This well-defined protocol maintains the consistency of handling input images and generating text outputs in a stable manner. The system developed enables the production of coherent and understandable image descriptions, supporting transparency in decision support, and has been developed based on the research goal to enable demonstrations of multimodal model

3.7 System Implement

The system phase of implementation includes the integration of a multimodal inference pipeline with a simulated PACS-like workflow. The main purpose of this phase is to show how the proposed AI-driven interpretation system works in clinical practice in a radiology environment. A web-based frontend is built by applying to Streamlit or Gradio, which include functionalities of uploading chest X-ray images, visualizing the corresponding interpretation, and generating textual outputs produced by AI. Its web-based interface is similar to a basic PACS viewer, providing users with the capability of viewing chest X-rays with generated reports side by side. The backend services are implemented via FastAPI, which handles the reception of images,

preprocessing, and running model inferences, as well as sending back the results to the frontend. System modularity, maintainability, and scalability are guaranteed thanks to the fact that frontend and backend components work independently of each other. For standard medical image handling, the `pydicom` library is used, and it adopts the visualization layout of the popular OHIF Viewer in clinical applications. MongoDB is used to store the uploaded images and generated textual outputs and is for testing purposes only, and no real patient data are involved. To ensure portability and reproducibility, the entire system is containerized with Docker. Even without full integrations with hospital systems such as the RIS or competing standards such as HL7, our prototype already well simulates key functions of a PACS uploading

3.8 Testing and Evaluation

The prototype testing and evaluation stage involves the determination of the technical functionality, system performance, and integration of the proposed system workflow. The major aim of this stage is to ensure that the developed prototype works in a correct and reliable manner as per its design. The representative chest X-ray images of the selected data are used to do technical functionality testing by self-testing to make sure that the whole system is working as planned. This involves ensuring that the system can upload images without crashing, run images through the trained inference pipeline, and produce text-based outputs without crashing. The consistency of the system with repeated image uploads and the inference execution is also monitored to make sure that the system will be capable of running consistently and without crashing. System performance is measured based on the response time taken to produce the outputs, and it is also measured based on the consistency in the textual results generated when used in a series of inferences. The reliability and operational stability of the multimodal inference pipeline are tested with the help of these tests. Workflow integration will be measured by looking at the functionality of the prototype in providing a PACS-style workflow, which includes image uploading, image visualization, and report presentation within a single interface. The assessment will be made on the system workflow coherence and the technical viability of deploying multimodal AI models to

a PACS-like setting. Results of this stage include results of technical functionality tests, results of system performance observation, and results of workflow integration, which will all present the viability and strength of the suggested prototype system.

3.9 Documentation

The documentation phase targeting on combining all methodological explanation discussions into a complete Final Year Project report. This ensures that the research process is transparent, reproducible, and compliant with academic standards.

REFERENCES

- Hopstaken, R. M., Witbraad, T., van Engelshoven, J. M. A., & Dinant, G. J. (1996). Interobserver variation in the interpretation of chest radiographs for pneumonia in community-acquired lower respiratory tract infections. *European Journal of Radiology*, 29(1), 29–35. [https://doi.org/10.1016/0720-048X\(95\)00716-3](https://doi.org/10.1016/0720-048X(95)00716-3)
- Stec, N., Arje, D., Moody, A. R., Krupinski, E. A., & Tyrrell, P. N. (2018). *A systematic review of fatigue in radiology: Is it a problem?* *American Journal of Roentgenology*, 210(4), 799–806. <https://doi.org/10.2214/AJR.17.18613>
- World Health Organization. (2023). *Global health observatory data: Diagnostic imaging and radiology*. Retrieved from <https://www.who.int/data/gho>
- He, Z., Wong, A. N. N., & Yoo, J. S. (2025). *Radiology report generation using automatic keyword adaptation, frequency-based multi-label classification and text-to-text large language models*. *Computers in Biology and Medicine*, 196, 110625. <https://doi.org/10.1016/j.compbiomed.2025.110625>
- Pham, T.-T., Brecheisen, J., Wu, C. C., Nguyen, H., Deng, Z., Adjero, D., Doretto, G., Choudhary, A., & Le, N. (2025). *ItpCtrl-AI: End-to-end interpretable and controllable artificial intelligence by modeling radiologists' intentions*. *Artificial Intelligence in Medicine*, 160, 103054. <https://www.sciencedirect.com/science/article/pii/S0933365724002963>.
- Thawakar, O., Shaker, A., Mullappilly, S. S., Cholakkal, H., Anwer, R. M., Khan, S., Laaksonen, J., & Khan, F. S. (2024). *XrayGPT: Chest radiographs summarization using large medical vision-language models*. *Proceedings of the 23rd Workshop on Biomedical Language Processing*, 440–448. <https://aclanthology.org/2024.bionlp-1.35.pdf>

- Huemann, Z., Tie, X., Hu, J., & Bradshaw, T. J. (2024). ConTEXTual Net: A multimodal vision-language model for segmentation of pneumothorax. *Journal of Imaging Informatics in Medicine*, 37, 1652–1663. <https://doi.org/10.1007/s10278-024-01051-8>.
- Li, X., Li, L., Jiang, Y., Wang, H., Qiao, X., Feng, T., Luo, H., & Zhao, Y. (2025). Vision-language models in medical image analysis: From simple fusion to general large models. *Information Fusion*, 118, 102995. <https://doi.org/10.1016/j.inffus.2025.102995>.
- Lu, Y., & Wang, A. (2025). Integrating language into medical visual recognition and reasoning: A survey. *Medical Image Analysis*, 102, 103514. <https://doi.org/10.1016/j.media.2025.103514>
- Ryu, J. S., Kang, H., Chu, Y., & Yang, S. (2025). Vision-language foundation models for medical imaging: A review of current practices and innovations. *Biomedical Engineering Letters*, 15, 809–830. <https://doi.org/10.1007/s13534-025-00484-6>.
- Sun, Y., Wen, X., Zhang, Y., Jin, L., Yang, C., Zhang, Q., Jiang, M., Xu, Z., Guo, W., Su, J., & Jiang, X. (2025). Visual-language foundation models in medical imaging: A systematic review and meta-analysis of diagnostic and analytical applications. *Computer Methods and Programs in Biomedicine*, 268, 108870. <https://doi.org/10.1016/j.cmpb.2025.108870>.
- Tanno, R., Barrett, D. G. T., Sellergren, A., Ghaisas, S., Dathathri, S., See, A., Welbl, J., Lau, C., Tu, T., Azizi, S., Singhal, K., Schaekermann, M., May, R., Lee, R., Man, S., Mahdavi, S., Ahmed, Z., Matias, Y., Barral, J., Eslami, S. M. A., Belgrave, D., Liu, Y., Kalidindi, S. R., Shetty, S., Natarajan, V., Kohli, P., Huang, P.-S., Karthikesalingam, A., & Ktena, I. (2024). Collaboration between clinicians and vision–language models in radiology report generation. *Nature Medicine*, 30, 3302–3321. <https://doi.org/10.1038/s41591-024-03302-1>.

- Wang, X., Peng, Y., Lu, L., Lu, Z., Bagheri, M., & Summers, R. M. (2017). ChestX-ray8: Hospital-scale chest X-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 2097–2106).
<https://doi.org/10.1109/CVPR.2017.369>.
- Englebert, A., Collin, A.-S., Cornu, O., & De Vleeschouwer, C. (2024). *Self-supervised vision-language alignment of deep learning representations for bone X-rays analysis*. arXiv.
<https://doi.org/10.48550/arXiv.2405.08932>.
- Islam, M. R., Hossain, M. Z., Ahmed, M., & Samu, M. S. (2025). *Vision-language models for automated chest X-ray interpretation: Leveraging ViT and GPT-2*. arXiv.
<https://doi.org/10.48550/arXiv.2501.12356>.
- Ng, C., Sun, L., & Tang, S. (2025). *X-Ray-CoT: Interpretable chest X-ray diagnosis with vision-language models via chain-of-thought reasoning*. arXiv.
<https://doi.org/10.48550/arXiv.2508.12455>.
- Wang, Y., Wang, Z., Zhu, D., Zhong, J., & Li, W. (2025). *Governance-ready small language models for medical imaging: Prompting, abstention, and PACS integration*. arXiv.
<https://doi.org/10.48550/arXiv.2508.133>.
- Joy, M. (2024). *AI integration in PACS: Advancing medical imaging and healthcare delivery*. International Journal of Scientific and Advanced Technology, 16(1), 2930.
<https://doi.org/10.71097/IJSAT.v16.i1.2930>
- Demner-Fushman, D., Kohli, M. D., Rosenman, M. B., Shooshan, S. E., Rodriguez, L., Antani, S., Thoma, G. R., & McDonald, C. J. (2016). Preparing a collection of radiology examinations for distribution and retrieval. *Journal of the American Medical Informatics*

Association, 23(2), 304–312. [Preparing a collection of radiology examinations for distribution and retrieval - PubMed](#).

Nguyen, N. H., et al. (2021). Deployment and validation of an AI system for chest X-ray interpretation. *arXiv preprint*.

<https://arxiv.org/abs/2104.02256>

Imaging the Chest. (2001). *Radiographic densities on chest radiographs*.

<https://x-raylady.com/downloadfile/imagingthechest.pdf>

Arora, D., & Mehta, Y. (2014). *Use of picture archiving and communication system for imaging of radiological films in cardiac surgical intensive care unit*. *Journal of Anaesthesiology Clinical Pharmacology*, 30(3), 447–448. <https://doi.org/10.4103/0970-9185.137306>.